

Theoretical Computer Science Exam, July 10, 2024

The exam consists of **4 exercises**. Available time: 1 hr 30 min.; you may write your answers in Italian, if you wish.

All exercises must be addressed for the written exam to be considered sufficient.

F.NAME L.NAME PERSON CODE

Exercise 1 (8 pts)

Consider the following language

$$L = \{ a^i b^j \mid i = 2j + 1 \wedge j > 0 \}$$

For instance $aaab, a^5b^2 \in L$, $aab, aabb \notin L$.

1. Design a minimal power automaton that accepts language L . Write a computation of the automaton that accepts string a^5b^2 and one that rejects string aab .
2. Design a grammar, of the least general type, that generates language L . Write a derivation in the grammar for the string $a^5b^2 \in L$.

Exercise 2 (8 pts)

Consider a subprogram $allDiff(in, nIn, k, out)$ whose parameters have the following meaning

- in is an input array of integer numbers that should include at least 3 elements
- nIn is the input actual number of elements of array in (notice that indices inside array in start from 1)
- k is an input integer value, assumed to be greater than 1
- out is a Boolean output, that is a Boolean result returned to the caller

In implementing the subprogram one can assume that all the elements of array in are strictly positive.

After execution of the subprogram the output parameter out must be true if and only if for every value stored in the array in there is another value, also stored in the array, such that the absolute difference between the two values is equal to k .

Example: a correct set of parameters is the following: $in = [12, 8, 3, 17]$, $nIn = 4$, $k = 5$, $out = \text{true}$.

- a) Write mathematical logic formulas for the pre- and post-conditions of program $allDiff$
- b) For each of the following sets of values of the parameters of the $allDiff$ subprogram, say if the precondition and postcondition are satisfied, and if the parameter values can be used to argue that the implementation of the subprogram is incorrect with respect to its specification by means of the pre- and post-condition. Shortly justify your answer.

1) $in = [4, 9, 6, 7]$, $nIn = 4$, $k = 5$, $out = \text{false}$

Precondition satisfied (Y/N): ...

Postcondition satisfied (Y/N): ...

Parameter values show that implementation is incorrect (Y/N): ...

Short justification: ...

2) $in = [4, 9, 11, 6]$, $nIn = 4$, $k = 5$, $out = \text{false}$

Precondition satisfied (Y/N): ...

Postcondition satisfied (Y/N): ...

Parameter values show that implementation is incorrect (Y/N): ...

Short justification: ...

3) $in = [4, 5, 7, 6]$, $nIn = 4$, $k = 1$, $out = \text{false}$

Precondition satisfied (Y/N): ...

Postcondition satisfied (Y/N): ...

Parameter values show that implementation is incorrect (Y/N): ...

Short justification: ...

Exercise 3 (8 pts)

Please answer the following questions; provide suitable concise justifications for your answers.

1. Is the set $S1 = \{ \langle x, y \rangle \mid x+y \text{ is a prime number} \}$ decidable? Is it semidecidable?
2. Is the set $S2 = \{ z \mid \forall x \forall y (f_z(x, y) = 1 \text{ if } x+y \text{ is a prime number; } f_z(x, y) = 0 \text{ otherwise}) \}$ decidable?
3. Is the set $S3 = \{ z \mid \exists x \exists y (f_z(x, y) = 1 \text{ if } x+y \text{ is a prime number; } f_z(x, y) = 0 \text{ otherwise}) \}$ decidable? Is it semidecidable?
4. Is the set $S4 = \{ \langle x, y \rangle \mid \forall z (f_z(x, y) = 1 \text{ if } x+y \text{ is a prime number; } f_z(x, y) = 0 \text{ otherwise}) \}$ decidable? Is it semidecidable?

Exercise 4 (8 pts)

For a fixed integer k , consider an alphabet $\Sigma = \{a_1, a_2, \dots, a_k\}$ (so that $|\Sigma|=k$) and the language L , over the alphabet Σ , defined as follows.

$$L = \{ x \in \Sigma^* \mid \text{no alphabet symbol occurs in } x \text{ more than } k \text{ times} \}$$

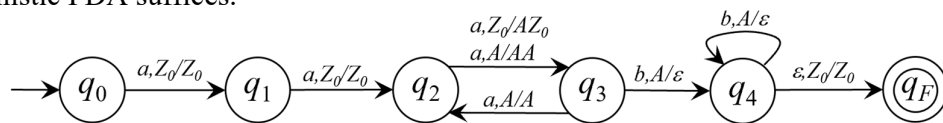
Describe in a clear and unambiguous way a Turing machine and a RAM machine that accept language L .

For both the Turing machine and the RAM evaluate the time and space complexity adopting all the available cost criteria.

Solutions

Exercise 1

1. Design a minimal power automaton that accepts language L . Write a computation of the automaton that accepts string a^5b^2 and one that rejects string aab .
A deterministic PDA suffices.



$\langle q_0, aaaaabb, Z_0 \rangle \vdash \langle q_1, aaaabb, Z_0 \rangle \vdash \langle q_2, aaabb, Z_0 \rangle \vdash \langle q_3, aabb, AZ_0 \rangle \vdash \langle q_2, abb, AZ_0 \rangle \vdash$
 $\vdash \langle q_3, bb, AAZ_0 \rangle \vdash \langle q_4, b, AZ_0 \rangle \vdash \langle q_4, \epsilon, Z_0 \rangle \vdash \langle q_F, \epsilon, Z_0 \rangle$ accept
 $\langle q_0, aab, Z_0 \rangle \vdash \langle q_1, ab, Z_0 \rangle \vdash \langle q_2, b, Z_0 \rangle$ reject

2. Design a grammar, of the least general type, that generates language L . Write a derivation in the grammar for the string $a^5b^2 \in L$.

$$S \rightarrow aA$$

$$A \rightarrow aAAb \mid aab$$

$$S \Rightarrow aA \Rightarrow aAAb \Rightarrow aAAbAb$$

Exercise 2

- a) Write mathematical logic formulas for the pre- and post-conditions of program *allDiff*

Precondition: $k > 1 \wedge nIn \geq 3 \wedge \forall i (1 \leq i \leq nIn \rightarrow in[i] > 0)$

Post condition: $out \leftrightarrow \forall i (1 \leq i \leq nIn \rightarrow \exists j (1 \leq j \leq nIn \wedge |in[i] - in[j]| = k))$

b) For each of the following sets of values of the parameters of the *allDiff* subprogram, say if the precondition and postcondition are satisfied, and if the parameter values can be used to argue that the implementation of the subprogram is incorrect with respect to its specification by means of the pre- and post-condition. Shortly justify your answer.

1) $in = [4, 9, 6, 7]$, $nIn = 4$, $k = 5$, $out = false$

Precondition satisfied (Y/N): yes

Postcondition satisfied (Y/N): yes

Parameter values show that implementation is incorrect (Y/N): no

Short justification: precondition and postcondition are both true

2) $in = [4, 9, 11, 6]$, $nIn = 4$, $k = 5$, $out = false$

Precondition satisfied (Y/N): yes

Postcondition satisfied (Y/N): no

Parameter values show that implementation is incorrect (Y/N): yes

Short justification: precondition is satisfied, postcondition is not, hence implementation is incorrect

3) $in = [4, 5, 7, 6]$, $nIn = 4$, $k = 1$, $out = false$

Precondition satisfied (Y/N): no, because $k=1$

Postcondition satisfied (Y/N): no (out should be true)

Parameter values show that implementation is incorrect (Y/N): no

Short justification: postcondition is not satisfied but precondition is also not satisfied, hence one cannot conclude that the implementation is incorrect

Exercise 3

1. Is the set $S1 = \{ \langle x, y \rangle \mid x+y \text{ is a prime number} \}$ decidable? Is it semidecidable?

$S1$ is decidable (and hence semidecidable); there exists a procedure that, given any pair of numbers $\langle x, y \rangle$, determines if $x+y$ is a prime number.

2. Is the set $S2 = \{ z \mid \forall x \forall y (f_z(x, y) = 1 \text{ if } x+y \text{ is a prime number; } f_z(x, y) = 0 \text{ otherwise}) \}$ decidable?

$S2$ is the set of indices (Goedel numbers) of the Turing machines that compute a function like the one mentioned in the answer concerning set $S1$. $S2$ satisfies the hypotheses of the Rice theorem, therefore it is not decidable.

3. Is the set $S3 = \{ z \mid \exists x \exists y (f_z(x, y) = 1 \text{ if } x+y \text{ is a prime number; } f_z(x, y) = 0 \text{ otherwise}) \}$ decidable? Is it semidecidable?

$S3$ is the set of indices (Goedel numbers) of the Turing machines that, for at least one pair of numbers $\langle x, y \rangle$, determines if $x+y$ is a prime number. $S3$ satisfies the hypotheses of the Rice theorem, therefore it is not decidable. $S3$ is semidecidable, by means of dovetail simulation (given a value z simulate the execution of f_z repeatedly for all possible arguments and for an increasing number of steps, etc. ...).

4. Is the set $S4 = \{ \langle x, y \rangle \mid \forall z (f_z(x, y) = 1 \text{ if } x+y \text{ is a prime number; } f_z(x, y) = 0 \text{ otherwise}) \}$ decidable? Is it semidecidable?

S4 is the empty set, because for no pair of values $\langle x, y \rangle$ all the computable functions applied to $\langle x, y \rangle$ return 1 if $x+y$ is a prime number and 0 otherwise. Therefore S4 is decidable and semidecidable.

Exercise 4

Since no alphabet symbol can occur more than k times in a string $x \in L$, the string length is bounded: in fact, in any string x such that $|x| > k^2$ at least one element of the alphabet occurs a number of times $> k$. Therefore, $\forall x \in L |x| \leq k^2$. The language L is finite, it can be accepted by a finite state automaton, hence by a 1-tape TM and by a RAM, and any computation will terminate in $\leq k^2$ steps. Therefore for the 1-tape TM: $T(n) \in \Theta(1)$ and $S(n) \in \Theta(1)$.

For the RAM, with both constant and logarithmic cost criteria: $T(n) \in \Theta(1)$, $S(n) \in \Theta(1)$.